

YOLO

March 16, 2026

0.0.1 Street View Blurring System YOLOv8

```
[1]: %pip install ultralytics
```

```
Requirement already satisfied: ultralytics in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages
(8.4.11)
Requirement already satisfied: numpy>=1.23.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (3.9.4)
Requirement already satisfied: opencv-python>=4.6.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (4.13.0.90)
Requirement already satisfied: pillow>=7.1.2 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (11.2.1)
Requirement already satisfied: pyyaml>=5.3.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (6.0.2)
Requirement already satisfied: requests>=2.23.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (1.13.1)
Requirement already satisfied: torch>=1.8.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (2.8.0)
Requirement already satisfied: torchvision>=0.9.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (0.23.0)
Requirement already satisfied: psutil>=5.8.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralytics) (7.2.2)
Requirement already satisfied: polars>=0.20.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
```

ultralitics) (1.36.1)
Requirement already satisfied: ultralytics-thop>=2.0.18 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
ultralitics) (2.0.18)
Requirement already satisfied: contourpy>=1.0.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (1.3.0)
Requirement already satisfied: cycler>=0.10 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (4.58.4)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (1.4.7)
Requirement already satisfied: packaging>=20.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (26.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (2.9.0.post0)
Requirement already satisfied: importlib-resources>=3.2.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
matplotlib>=3.3.0->ultralitics) (6.5.2)
Requirement already satisfied: zipp>=3.1.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
importlib-resources>=3.2.0->matplotlib>=3.3.0->ultralitics) (3.23.0)
Requirement already satisfied: polars-runtime-32==1.36.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
polars>=0.20.0->ultralitics) (1.36.1)
Requirement already satisfied: six>=1.5 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
python-dateutil>=2.7->matplotlib>=3.3.0->ultralitics) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
requests>=2.23.0->ultralitics) (3.4.2)
Requirement already satisfied: idna<4,>=2.5 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
requests>=2.23.0->ultralitics) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
requests>=2.23.0->ultralitics) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from

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requests>=2.23.0->ultralitics) (2025.6.15)
Requirement already satisfied: filelock in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
torch>=1.8.0->ultralitics) (3.19.1)
Requirement already satisfied: typing-extensions>=4.10.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
torch>=1.8.0->ultralitics) (4.15.0)
Requirement already satisfied: sympy>=1.13.3 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
torch>=1.8.0->ultralitics) (1.14.0)
Requirement already satisfied: networkx in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
torch>=1.8.0->ultralitics) (3.2.1)
Requirement already satisfied: jinja2 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
torch>=1.8.0->ultralitics) (3.1.6)
Requirement already satisfied: fsspec in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
torch>=1.8.0->ultralitics) (2025.10.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
sympy>=1.13.3->torch>=1.8.0->ultralitics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
jinja2->torch>=1.8.0->ultralitics) (3.0.2)
Note: you may need to restart the kernel to use updated packages.

```

```

[1]: import sys
      !{sys.executable} -m pip install --upgrade --force-reinstall nbformat

```

```

Collecting nbformat
  Using cached nbformat-5.10.4-py3-none-any.whl.metadata (3.6 kB)
Collecting fastjsonschema>=2.15 (from nbformat)
  Using cached fastjsonschema-2.21.2-py3-none-any.whl.metadata (2.3 kB)
Collecting jsonschema>=2.6 (from nbformat)
  Downloading jsonschema-4.25.1-py3-none-any.whl.metadata (7.6 kB)
Collecting jupyter-core!=5.0.*,>=4.12 (from nbformat)
  Downloading jupyter_core-5.8.1-py3-none-any.whl.metadata (1.6 kB)
Collecting traitlets>=5.1 (from nbformat)
  Using cached traitlets-5.14.3-py3-none-any.whl.metadata (10 kB)
Collecting attrs>=22.2.0 (from jsonschema>=2.6->nbformat)
  Using cached attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Collecting jsonschema-specifications>=2023.03.6 (from jsonschema>=2.6->nbformat)
  Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9
kB)
Collecting referencing>=0.28.4 (from jsonschema>=2.6->nbformat)
  Downloading referencing-0.36.2-py3-none-any.whl.metadata (2.8 kB)
Collecting rpds-py>=0.7.1 (from jsonschema>=2.6->nbformat)

```

```

    Downloading rpds_py-0.27.1-cp39-cp39-macosx_11_0_arm64.whl.metadata (4.2 kB)
Collecting platformdirs>=2.5 (from jupyter-core!=5.0.*,>=4.12->nbformat)
    Downloading platformdirs-4.4.0-py3-none-any.whl.metadata (12 kB)
Collecting typing-extensions>=4.4.0 (from
referencing>=0.28.4->jsonschema>=2.6->nbformat)
    Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached nbformat-5.10.4-py3-none-any.whl (78 kB)
Using cached fastjsonschema-2.21.2-py3-none-any.whl (24 kB)
Downloading jsonschema-4.25.1-py3-none-any.whl (90 kB)
Using cached attrs-25.4.0-py3-none-any.whl (67 kB)
Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl (18 kB)
Downloading jupyter_core-5.8.1-py3-none-any.whl (28 kB)
Downloading platformdirs-4.4.0-py3-none-any.whl (18 kB)
Downloading referencing-0.36.2-py3-none-any.whl (26 kB)
Downloading rpds_py-0.27.1-cp39-cp39-macosx_11_0_arm64.whl (354 kB)
Using cached traitlets-5.14.3-py3-none-any.whl (85 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: fastjsonschema, typing-extensions, traitlets,
rpds-py, platformdirs, attrs, referencing, jupyter-core, jsonschema-
specifications, jsonschema, nbformat
Attempting uninstall: fastjsonschema
    Found existing installation: fastjsonschema 2.21.1
    Uninstalling fastjsonschema-2.21.1:
        Successfully uninstalled fastjsonschema-2.21.1
Attempting uninstall: typing-extensions
    Found existing installation: typing_extensions 4.15.0
    Uninstalling typing_extensions-4.15.0:
        Successfully uninstalled typing_extensions-4.15.0
1/11 [typing-extensions]
    Attempting uninstall: traitlets
1/11 [typing-extensions]
    Found existing installation: traitlets 5.14.3
1/11 [typing-extensions]
    Uninstalling traitlets-5.14.3:
1/11 [typing-extensions]
    Successfully uninstalled traitlets-5.14.3
1/11 [typing-extensions]
    Attempting uninstall: rpds-py
1/11 [typing-extensions]
    Found existing installation: rpds-py 0.25.1
1/11 [typing-extensions]
    Uninstalling rpds-py-0.25.1:
1/11 [typing-extensions]

```

```

        Successfully uninstalled rpds-py-0.25.1
1/11 [typing-extensions]
    Attempting uninstall: platformdirs
1/11 [typing-extensions]
    Found existing installation: platformdirs 4.4.0
1/11 [typing-extensions]
    Uninstalling platformdirs-4.4.0:
1/11 [typing-extensions]
    Successfully uninstalled platformdirs-4.4.0
1/11 [typing-extensions]
    Attempting uninstall: attrs
1/11 [typing-extensions]
    Found existing installation: attrs 25.3.0
1/11 [typing-extensions]
    Uninstalling attrs-25.3.0:
1/11 [typing-extensions]
    Successfully uninstalled attrs-25.3.0
1/11 [typing-extensions]
    Attempting uninstall: referencing
1/11 [typing-extensions]
    Found existing installation: referencing 0.36.2
1/11 [typing-extensions]
    Uninstalling referencing-0.36.2:
1/11 [typing-extensions]
    Successfully uninstalled referencing-0.36.2
1/11 [typing-extensions]
    Attempting uninstall: jupyter-core
1/11 [typing-extensions]
    Found existing installation: jupyter_core 5.8.1
1/11 [typing-extensions]
    Uninstalling jupyter_core-5.8.1:
1/11 [typing-extensions]
    Successfully uninstalled jupyter_core-5.8.190m
7/11 [jupyter-core]
    Attempting uninstall: jsonschema-specifications
7/11 [jupyter-core]

```

```

Found existing installation: jsonschema-specifications 2025.4.1
7/11 [jupyter-core]
Uninstalling jsonschema-specifications-2025.4.1:
7/11 [jupyter-core]
Successfully uninstalled jsonschema-specifications-2025.4.1m
7/11 [jupyter-core]
Attempting uninstall: jsonschema[90m
7/11 [jupyter-core]
Found existing installation: jsonschema 4.24.0
7/11 [jupyter-core]
Uninstalling jsonschema-4.24.0:90m
7/11 [jupyter-core]
Successfully uninstalled jsonschema-4.24.0
7/11 [jupyter-core]
Attempting uninstall: nbformat
9/11 [jsonschema]
Found existing installation: nbformat 5.10.40m
9/11 [jsonschema]
Uninstalling nbformat-5.10.4:
9/11 [jsonschema]
Successfully uninstalled nbformat-5.10.4
9/11 [jsonschema]
11/11
[nbformat][0m [jsonschema]
Successfully installed attrs-25.4.0 fastjsonschema-2.21.2
jsonschema-4.25.1 jsonschema-specifications-2025.9.1 jupyter-core-5.8.1
nbformat-5.10.4 platformdirs-4.4.0 referencing-0.36.2 rpds-py-0.27.1
traitlets-5.14.3 typing-extensions-4.15.0

```

```
[2]: %pip install opencv-python
```

```

Requirement already satisfied: opencv-python in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages
(4.13.0.90)
Requirement already satisfied: numpy>=2 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
opencv-python) (2.0.2)
Note: you may need to restart the kernel to use updated packages.

```

```
[3]: import ultralytics
import cv2
from ultralytics import YOLO
```

```
# Verify system setup for training
ultralalytics.checks()
```

Ultralytics 8.4.11 Python-3.9.6 torch-2.8.0 CPU (Apple M5 Pro)
Setup complete (15 CPUs, 24.0 GB RAM, 95.9/926.3 GB disk)

```
[4]: from pathlib import Path

base_path = Path("./data")

def check_dataset_consistency(base):
    splits = ['train', 'val', 'test']

    print(f"{'Split':<10} | {'Images':<10} | {'Labels':<10} | {'Status'}")
    print("-" * 50)

    for split in splits:
        # Correct pathing based on your screenshot:
        # Data/images/train/ and Data/labels/train/
        img_folder = base / 'images' / split
        lbl_folder = base / 'labels' / split

        # Count files (handling case sensitivity and common extensions)
        img_extensions = ['*.jpg', '*.jpeg', '*.png', '*.JPG', '*.PNG']
        img_count = sum(len(list(img_folder.glob(ext))) for ext in
↳img_extensions)
        lbl_count = len(list(lbl_folder.glob('*.txt')))

        status = " Match" if img_count == lbl_count and img_count > 0 else "
↳Mismatch or Empty"
        print(f"{'split':<10} | {'img_count':<10} | {'lbl_count':<10} | {'status'}")

    check_dataset_consistency(base_path)
```

Split	Images	Labels	Status
train	23604	23604	Match
val	1073	1073	Match
test	386	386	Match

```
[5]: #converting YOLO coordinates to pixel coordinates

import cv2
import matplotlib.pyplot as plt

def audit_sample(image_path, label_path):
    # Load image
```

```

img = cv2.imread(str(image_path))
h, w, _ = img.shape

# Read YOLO label (class, cx, cy, bw, bh)
with open(label_path, 'r') as f:
    lines = f.readlines()

for line in lines:
    data = line.split()
    cx, cy, bw, bh = map(float, data[1:])

    # BUSINESS LOGIC: Convert normalized to pixel coordinates
    x1 = int((cx - bw/2) * w)
    y1 = int((cy - bh/2) * h)
    x2 = int((cx + bw/2) * w)
    y2 = int((cy + bh/2) * h)

    # Draw green bounding box
    cv2.rectangle(img, (x1, y1), (x2, y2), (0, 255, 0), 2)
    cv2.putText(img, "License Plate", (x1, y1-10),
                cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

# Display
plt.figure(figsize=(10, 6))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Data Audit: Bounding Box Verification")
plt.axis('off')
plt.show()

# Test on one image from your 25,470 training set
# Update filenames to real ones in your folder
# audit_sample(data_splits['train']/images/example.jpg',
#               data_splits['train']/labels/example.txt')

```

0.1 Creating config yaml

```

[7]: import yaml

# Use .resolve() to get the full absolute path on your Mac
abs_base = base_path.resolve()

yaml_content = {
    'train': str(abs_base / 'images' / 'train'),
    'val': str(abs_base / 'images' / 'val'),
    'test': str(abs_base / 'images' / 'test'),
    'nc': 1,
    'names': ['license_plate']
}

```

```

}

with open('license_plate.yaml', 'w') as f:
    yaml.dump(yaml_content, f)

print(" license_plate.yaml created successfully.")

```

license_plate.yaml created successfully.

0.2 Training phase

```

[8]: from ultralytics import YOLO

# 1. Load the Model
# We use 'yolov8n.pt' (Nano) because it balances speed and accuracy
# and is less likely to overfit on our reduced dataset of ~2,100 images.
model = YOLO('yolov8n.pt')

# 2. Train the Model
results = model.train(
    data='license_plate.yaml',      # The config file we created earlier
    epochs=50,                      # 50 iterations to allow the model to learn
    features                         # ↳ features
    imgsiz=640,                     # Resize images to 640x640 for consistency
    batch=16,                       # Process 16 images at a time
    device='mps',                   # Change to 'cpu' if not on Apple Silicon
    name='plate_anonymizer_v1'      # Folder name for your logs and weights
)

```

/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: <https://github.com/urllib3/urllib3/issues/3020>

warnings.warn(

New <https://pypi.org/project/ultralytics/8.4.21> available Update with 'pip install -U ultralytics'

Ultralytics 8.4.11 Python-3.9.6 torch-2.8.0 MPS (Apple M5 Pro)

engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=license_plate.yaml, degrees=0.0, deterministic=True, device=mps, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=50, erasing=0.4, exist_ok=False,fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsiz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train,

```

model=yolov8n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=plate_anonymizer_v13, nbs=64, nms=False, opset=None, optimize=False,
optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False, project=None, rect=False,
resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False,
save_crop=False, save_dir=/Users/kishankunal/workspace/AppliedAI/runs/detect/pla
te_anonymizer_v13, save_frames=False, save_json=False, save_period=-1,
save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True,
show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None,
split=val, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml,
translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False,
warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005,
workers=8, workspace=None
Overriding model.yaml nc=80 with nc=1

```

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]				
2	-1	1	7360	ultralytics.nn.modules.block.C2f
[32, 32, 1, True]				
3	-1	1	18560	ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]				
4	-1	2	49664	ultralytics.nn.modules.block.C2f
[64, 64, 2, True]				
5	-1	1	73984	ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]				
6	-1	2	197632	ultralytics.nn.modules.block.C2f
[128, 128, 2, True]				
7	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
8	-1	1	460288	ultralytics.nn.modules.block.C2f
[256, 256, 1, True]				
9	-1	1	164608	ultralytics.nn.modules.block.SPPF
[256, 256, 5]				
10	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
11	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
12	-1	1	148224	ultralytics.nn.modules.block.C2f
[384, 128, 1]				
13	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
14	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
15	-1	1	37248	ultralytics.nn.modules.block.C2f

```

[192, 64, 1]
 16          -1  1      36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
 17      [-1, 12]  1          0  ultralytics.nn.modules.conv.Concat
[1]
 18          -1  1      123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
 19          -1  1      147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 20      [-1, 9]  1          0  ultralytics.nn.modules.conv.Concat
[1]
 21          -1  1      493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
 22      [15, 18, 21]  1      751507  ultralytics.nn.modules.head.Detect
[1, 16, None, [64, 128, 256]]
Model summary: 130 layers, 3,011,043 parameters, 3,011,027 gradients, 8.2 GFLOPs

```

Transferred 319/355 items from pretrained weights

Freezing layer 'model.22.dfl.conv.weight'

train: Fast image access (ping: 0.0±0.0 ms, read: 178.9±129.3

MB/s, size: 63.8 KB)

train: Scanning

/Users/kishankunal/workspace/AppliedAI/YOLO8-License Plate Detection &
Blurring/data/labels/train... 23604 images, 14 backgrounds, 0 corrupt: 100%
23604/23604 5.8Kit/s 4.1s0.1s

train: New cache created:

/Users/kishankunal/workspace/AppliedAI/YOLO8-License Plate Detection &
Blurring/data/labels/train.cache

val: Fast image access (ping: 0.0±0.0 ms, read: 939.6±570.2 MB/s,
size: 357.2 KB)

val: Scanning

/Users/kishankunal/workspace/AppliedAI/YOLO8-License Plate Detection &
Blurring/data/labels/val... 1073 images, 0 backgrounds, 0 corrupt: 100%
1073/1073 5.9Kit/s 0.2s3s

val: New cache created:

/Users/kishankunal/workspace/AppliedAI/YOLO8-License Plate Detection &
Blurring/data/labels/val.cache

Plotting labels to /Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anon
ymizer_v13/labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and
'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'
automatically...

optimizer: MuSGD(lr=0.01, momentum=0.9) with parameter groups 57
weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)

Image sizes 640 train, 640 val

Using 0 dataloader workers

Logging results to

/Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anonymizer_v13

Starting training for 50 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/50	4.27G	1.253	1.358	1.14	11	
640: 100%	1476/1476	1.9it/s	12:47<0.6s			
WARNING NMS time limit 3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 3%	1/34	34.8s/it	10.4s<19:07	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 6%	2/34	15.6s/it	17.2s<8:18	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 9%	3/34	12.8s/it	26.3s<6:37	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 12%	4/34	11.5s/it	35.6s<5:46	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 15%	5/34	10.0s/it	43.3s<4:50	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 18%	6/34	8.6s/it	49.8s<4:02	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 21%	7/34	8.1s/it	57.0s<3:40	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 24%	8/34	8.0s/it	1:05<3:27	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	5.5s/it	3:09	3.6ss		
	all	1073	1573	0.694	0.554	0.579
0.269						

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/50	4.28G	1.257	0.7562	1.162	6	
640: 100%	1476/1476	1.7it/s	14:19<0.6s			
WARNING NMS time limit 3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 3%	1/34	26.3s/it	7.9s<14:29	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 6%	2/34	13.4s/it	14.2s<7:10	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 9%	3/34	11.1s/it	22.0s<5:43	WARNING	NMS time limit	
3.600s exceeded						

	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 12%	4/34	9.3s/it	28.7s<4:38	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 15%	5/34	8.7s/it	36.4s<4:13	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 18%	6/34	8.3s/it	43.8s<3:52	WARNING	NMS time limit	
3.600s exceeded						
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	3.8s/it	2:081.3ss			
	all	1073	1573	0.723	0.515	0.536
0.241						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
3/50	4.29G	1.287	0.7509	1.204	6	
640: 100%	1476/1476	1.8it/s	13:41<0.6s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	2.2s/it	1:152.0sss			
	all	1073	1573	0.566	0.555	0.519
0.227						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
4/50	4.29G	1.276	0.7298	1.205	9	
640: 100%	1476/1476	1.9it/s	13:04<0.6s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.2s/it	40.2s0.9ss			
	all	1073	1573	0.733	0.619	0.612
0.27						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
5/50	4.29G	1.238	0.6732	1.178	7	
640: 100%	1476/1476	2.0it/s	12:13<0.6s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.2s/it	40.2s0.7ss			
	all	1073	1573	0.76	0.615	0.67
0.306						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
6/50	4.29G	1.213	0.6382	1.165	3	
640: 100%	1476/1476	2.1it/s	11:43<0.5s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.0s/it	34.8s0.8ss			
	all	1073	1573	0.767	0.645	0.687
0.324						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
7/50	4.29G	1.2	0.6187	1.156	7	

640: 100% 1476/1476 2.1it/s 11:39<0.6s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.1it/s 32.4s0.7ss
 all 1073 1573 0.792 0.661 0.696
 0.326

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 8/50 4.29G 1.191 0.599 1.147 6
 640: 100% 1476/1476 2.1it/s 11:29<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.4it/s 23.7s0.6ss
 all 1073 1573 0.795 0.681 0.734
 0.35

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 9/50 4.29G 1.171 0.5811 1.138 6
 640: 100% 1476/1476 2.2it/s 11:11<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.0it/s 33.0s0.7ss
 all 1073 1573 0.792 0.697 0.74
 0.349

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 10/50 4.29G 1.166 0.5701 1.134 10
 640: 100% 1476/1476 2.2it/s 10:58<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.3it/s 25.7s0.6ss
 all 1073 1573 0.832 0.699 0.742
 0.359

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 11/50 4.29G 1.16 0.5651 1.129 9
 640: 100% 1476/1476 2.2it/s 10:58<0.6s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.4it/s 23.8s0.7ss
 all 1073 1573 0.823 0.725 0.767
 0.374

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 12/50 4.29G 1.15 0.5549 1.127 12
 640: 100% 1476/1476 2.2it/s 11:10<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.3it/s 26.8s0.6ss
 all 1073 1573 0.834 0.699 0.754
 0.372

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 13/50 4.29G 1.146 0.5502 1.122 5

640: 100% 1476/1476 2.2it/s 11:03<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.4it/s 24.8s0.6ss
 all 1073 1573 0.838 0.706 0.765
 0.386

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 14/50 4.29G 1.139 0.5425 1.122 7
 640: 100% 1476/1476 2.2it/s 11:08<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.6it/s 21.9s0.6ss
 all 1073 1573 0.834 0.719 0.77
 0.387

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 15/50 4.3G 1.134 0.5347 1.115 7
 640: 100% 1476/1476 2.3it/s 10:46<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.4it/s 24.3s0.6ss
 all 1073 1573 0.822 0.751 0.784
 0.398

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 16/50 4.29G 1.127 0.5251 1.112 9
 640: 100% 1476/1476 2.3it/s 10:49<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 20.4s0.6s
 all 1073 1573 0.846 0.758 0.799
 0.405

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 17/50 4.29G 1.123 0.5201 1.109 5
 640: 100% 1476/1476 2.3it/s 10:40<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.4it/s 24.8s0.6ss
 all 1073 1573 0.861 0.748 0.811
 0.413

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 18/50 4.3G 1.117 0.5147 1.106 7
 640: 100% 1476/1476 2.3it/s 10:44<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.6it/s 20.7s0.6ss
 all 1073 1573 0.841 0.766 0.811
 0.413

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 19/50 4.29G 1.111 0.5106 1.105 7

640: 100% 1476/1476 2.3it/s 10:40<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 19.5s0.6s
 all 1073 1573 0.858 0.748 0.806
 0.419

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 20/50 4.29G 1.111 0.5069 1.104 6
 640: 100% 1476/1476 2.3it/s 10:49<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.6it/s 21.1s0.6ss
 all 1073 1573 0.844 0.763 0.805
 0.417

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 21/50 4.29G 1.105 0.5011 1.099 5
 640: 100% 1476/1476 2.4it/s 10:25<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.0s0.6s
 all 1073 1573 0.84 0.76 0.808
 0.419

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 22/50 4.29G 1.102 0.4978 1.099 5
 640: 100% 1476/1476 2.3it/s 10:32<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.6it/s 20.6s0.5s
 all 1073 1573 0.851 0.77 0.821
 0.425

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 23/50 4.3G 1.099 0.4975 1.092 7
 640: 100% 1476/1476 2.3it/s 10:37<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.4s0.6s
 all 1073 1573 0.854 0.764 0.825
 0.43

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 24/50 4.3G 1.089 0.4899 1.091 7
 640: 100% 1476/1476 2.3it/s 10:35<0.6s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 20.6s0.6ss
 all 1073 1573 0.862 0.771 0.826
 0.434

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 25/50 4.3G 1.084 0.4848 1.086 10

640: 100% 1476/1476 2.3it/s 10:40<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 20.0s0.5ss
 all 1073 1573 0.847 0.781 0.826
 0.431

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 26/50 4.3G 1.083 0.4795 1.086 8
 640: 100% 1476/1476 2.4it/s 10:27<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 19.5s0.5s
 all 1073 1573 0.86 0.782 0.829
 0.433

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 27/50 4.3G 1.081 0.4783 1.086 9
 640: 100% 1476/1476 2.3it/s 10:36<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.4s0.5s
 all 1073 1573 0.857 0.777 0.833
 0.437

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 28/50 4.3G 1.08 0.4759 1.083 5
 640: 100% 1476/1476 2.3it/s 10:29<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.0s0.5s
 all 1073 1573 0.867 0.777 0.83
 0.437

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 29/50 4.29G 1.075 0.4703 1.08 6
 640: 100% 1476/1476 2.4it/s 10:21<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 19.5s0.6s
 all 1073 1573 0.853 0.78 0.831
 0.439

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 30/50 4.32G 1.073 0.4707 1.083 6
 640: 100% 1476/1476 2.3it/s 10:33<0.3s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.9it/s 18.3s0.5s
 all 1073 1573 0.864 0.772 0.832
 0.442

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 31/50 4.29G 1.059 0.4591 1.074 4

640: 100% 1476/1476 2.4it/s 10:20<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.9it/s 17.6s0.5s
 all 1073 1573 0.863 0.774 0.834
 0.443

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 32/50 4.3G 1.068 0.4608 1.076 9
 640: 100% 1476/1476 2.4it/s 10:26<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 18.9s0.5s
 all 1073 1573 0.859 0.778 0.835
 0.442

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 33/50 4.29G 1.057 0.4555 1.074 6
 640: 100% 1476/1476 2.4it/s 10:26<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.7it/s 19.8s0.6s
 all 1073 1573 0.859 0.785 0.839
 0.445

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 34/50 4.3G 1.055 0.4544 1.074 5
 640: 100% 1476/1476 2.4it/s 10:24<0.6s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.9it/s 18.3s0.5s
 all 1073 1573 0.855 0.787 0.842
 0.446

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 35/50 4.3G 1.047 0.4486 1.068 8
 640: 100% 1476/1476 2.4it/s 10:23<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 18.9s0.5s
 all 1073 1573 0.859 0.789 0.845
 0.448

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 36/50 4.3G 1.049 0.4463 1.069 7
 640: 100% 1476/1476 2.4it/s 10:26<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.3s0.5s
 all 1073 1573 0.853 0.797 0.848
 0.45

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 37/50 4.29G 1.039 0.4406 1.064 4

640: 100% 1476/1476 2.4it/s 10:22<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.0s0.6s
 all 1073 1573 0.851 0.794 0.847
 0.45

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 38/50 4.3G 1.035 0.4365 1.06 10
 640: 100% 1476/1476 2.4it/s 10:13<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.0s0.6s
 all 1073 1573 0.854 0.794 0.846
 0.45

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 39/50 4.3G 1.028 0.4324 1.06 6
 640: 100% 1476/1476 2.4it/s 10:19<0.5s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.9it/s 18.2s0.5s
 all 1073 1573 0.858 0.795 0.846
 0.451

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 40/50 4.3G 1.024 0.4312 1.057 6
 640: 100% 1476/1476 2.4it/s 10:10<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.9it/s 17.7s0.5s
 all 1073 1573 0.859 0.795 0.846
 0.451

Closing dataloader mosaic

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 41/50 4.29G 1.002 0.3817 1.082 4
 640: 100% 1476/1476 2.4it/s 10:23<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.9it/s 18.3s0.5s
 all 1073 1573 0.864 0.791 0.848
 0.451

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
 42/50 4.29G 0.995 0.3747 1.077 4
 640: 100% 1476/1476 2.4it/s 10:20<0.4s
 Class Images Instances Box(P R mAP50
 mAP50-95): 100% 34/34 1.8it/s 19.1s0.5s
 all 1073 1573 0.86 0.792 0.847
 0.452

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size

43/50	4.3G	0.986	0.3719	1.073	3	
640: 100%	1476/1476	2.4it/s	10:26<0.5s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.8it/s	18.6s0.6s			
	all	1073	1573	0.857	0.793	0.847
0.451						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
44/50	4.3G	0.9782	0.3661	1.073	6	
640: 100%	1476/1476	2.4it/s	10:24<0.5s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.8it/s	18.7s0.5s			
	all	1073	1573	0.862	0.79	0.844
0.451						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
45/50	4.29G	0.972	0.3626	1.065	4	
640: 100%	1476/1476	2.4it/s	10:19<0.5s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.9it/s	18.4s0.5s			
	all	1073	1573	0.859	0.792	0.845
0.451						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
46/50	4.29G	0.9629	0.3558	1.063	4	
640: 100%	1476/1476	2.4it/s	10:15<0.3s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.9it/s	17.7s0.5s			
	all	1073	1573	0.859	0.791	0.845
0.451						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
47/50	4.29G	0.9534	0.352	1.058	4	
640: 100%	1476/1476	2.4it/s	10:10<0.5s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.7it/s	20.2s0.5ss			
	all	1073	1573	0.86	0.791	0.845
0.452						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
48/50	4.29G	0.9484	0.347	1.055	4	
640: 100%	1476/1476	2.4it/s	10:22<0.5s			
	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	1.9it/s	18.2s0.5s			
	all	1073	1573	0.86	0.79	0.846
0.452						

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
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	49/50	4.29G	0.9421	0.3445	1.051	3	
640: 100%	1476/1476	2.4it/s	10:15<0.5s				
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95): 100%	34/34	1.8it/s	18.5s0.5s				
	all	1073	1573	0.861	0.788	0.845	
0.453							

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
50/50	4.29G	0.934	0.3411	1.047	4		
640: 100%	1476/1476	2.4it/s	10:19<0.5s				
	Class	Images	Instances	Box(P	R	mAP50	
mAP50-95): 100%	34/34	1.8it/s	18.6s0.5s				
	all	1073	1573	0.864	0.788	0.844	
0.453							

50 epochs completed in 9.464 hours.

Optimizer stripped from /Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anonymizer_v13/weights/last.pt, 6.2MB

Optimizer stripped from /Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anonymizer_v13/weights/best.pt, 6.2MB

Validating /Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anonymizer_v13/weights/best.pt...

Ultralytics 8.4.11 Python-3.9.6 torch-2.8.0 MPS (Apple M5 Pro)

Model summary (fused): 73 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs

	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	34/34	2.0s/it	1:081.0ss			
	all	1073	1573	0.861	0.789	0.845
0.453						

Speed: 0.3ms preprocess, 25.2ms inference, 0.0ms loss, 4.8ms postprocess per image

Results saved to

/Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anonymizer_v13

1 Evaluation & Performance Metrics

```
[2]: # This will generate a 'results.csv' and confusion matrix in 'runs/detect/val'
from ultralytics import YOLO

# Load your specific trained weights
# Replace the path below with the actual path to your best.pt
model = YOLO('/Users/kishankunal/workspace/AppliedAI/runs/detect/plate_anonymizer_v13/weights/best.pt')

# Now run your validation code
metrics = model.val(data='license_plate.yaml')
print(f"mAP@50-95: {metrics.box.map}")
```

Ultralytics 8.4.11 Python-3.9.6 torch-2.8.0 CPU (Apple M5 Pro)
Model summary (fused): 73 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (ping: 0.0±0.0 ms, read: 865.0±924.7 MB/s, size: 258.7 KB)
val: Scanning /Users/kishankunal/workspace/AppliedAI/YOLO8-LicensesPlateDetectionBlurring/data/labels/val.cache... 1073 images, 0 backgrounds, 0 corrupt: 100% 1073/1073 132.4Mit/s 0.0s

	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%		68/68	1.3s/it	1:281.3sss		
	all	1073	1573	0.867	0.786	0.844

0.453
Speed: 0.1ms preprocess, 77.4ms inference, 0.0ms loss, 0.1ms postprocess per image
Results saved to /Users/kishankunal/workspace/AppliedAI/runs/detect/val2
mAP@50-95: 0.45311050933547214

2 Post-Processing

```
[3]: import cv2
import os
from pathlib import Path

# Now your code will work:
test_images = list(Path("./data/images/test").glob("*.jpg"))

def process_and_blur(image_path, output_folder, model):
    img = cv2.imread(image_path)
    results = model(img)[0]

    for box in results.bboxes:
        # Confidence check as per your document
        if box.conf > 0.4:
            # Get pixel coordinates
            x1, y1, x2, y2 = map(int, box.xyxy[0])

            # Extract ROI (Region of Interest)
            plate_region = img[y1:y2, x1:x2]

            # Apply Gaussian Blur (Privacy Compliance)
            # Higher numbers in (k, k) = more blur
            blurred_plate = cv2.GaussianBlur(plate_region, (35, 35), 0)

            # Overwrite original area
            img[y1:y2, x1:x2] = blurred_plate

    # Save the redacted output
    filename = os.path.basename(image_path)
```

```

cv2.imwrite(os.path.join(output_folder, f"redacted_{filename}"), img)

# Apply to your TEST folder
test_images = list(Path("./data/images/test").glob("*.jpg"))
os.makedirs("output_redacted", exist_ok=True)

for img_p in test_images:
    process_and_blur(str(img_p), "output_redacted", model)

```

0: 576x640 1 license_plate, 22.9ms
Speed: 2.8ms preprocess, 22.9ms inference, 0.3ms postprocess per image at shape (1, 3, 576, 640)

0: 512x640 2 license_plates, 18.6ms
Speed: 1.1ms preprocess, 18.6ms inference, 0.3ms postprocess per image at shape (1, 3, 512, 640)

0: 480x640 1 license_plate, 16.7ms
Speed: 0.9ms preprocess, 16.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 17.8ms
Speed: 1.1ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 640x480 1 license_plate, 17.3ms
Speed: 1.1ms preprocess, 17.3ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 480x640 3 license_plates, 17.9ms
Speed: 1.0ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.7ms
Speed: 1.0ms preprocess, 18.7ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.3ms
Speed: 0.9ms preprocess, 17.3ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 19.3ms
Speed: 1.4ms preprocess, 19.3ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.4ms
Speed: 0.9ms preprocess, 17.4ms inference, 0.3ms postprocess per image at shape

(1, 3, 448, 640)

0: 448x640 3 license_plates, 16.2ms

Speed: 0.8ms preprocess, 16.2ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 19.2ms

Speed: 1.0ms preprocess, 19.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 20.0ms

Speed: 1.1ms preprocess, 20.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 20.3ms

Speed: 0.9ms preprocess, 20.3ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 20.5ms

Speed: 0.9ms preprocess, 20.5ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 38.4ms

Speed: 1.0ms preprocess, 38.4ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 3 license_plates, 24.4ms

Speed: 1.1ms preprocess, 24.4ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 23.2ms

Speed: 1.0ms preprocess, 23.2ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 544x640 1 license_plate, 25.2ms

Speed: 1.1ms preprocess, 25.2ms inference, 0.2ms postprocess per image at shape (1, 3, 544, 640)

0: 448x640 1 license_plate, 21.5ms

Speed: 0.9ms preprocess, 21.5ms inference, 0.5ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 23.0ms

Speed: 1.0ms preprocess, 23.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 19.8ms

Speed: 1.0ms preprocess, 19.8ms inference, 0.3ms postprocess per image at shape

(1, 3, 448, 640)

0: 640x480 2 license_plates, 23.7ms

Speed: 1.0ms preprocess, 23.7ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 480)

0: 480x640 1 license_plate, 21.6ms

Speed: 0.9ms preprocess, 21.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 24.1ms

Speed: 1.0ms preprocess, 24.1ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 384x640 1 license_plate, 16.4ms

Speed: 0.8ms preprocess, 16.4ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 448x640 1 license_plate, 20.6ms

Speed: 0.8ms preprocess, 20.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 23.4ms

Speed: 0.8ms preprocess, 23.4ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 2 license_plates, 29.5ms

Speed: 1.2ms preprocess, 29.5ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 22.6ms

Speed: 1.0ms preprocess, 22.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 20.4ms

Speed: 0.9ms preprocess, 20.4ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 19.4ms

Speed: 0.8ms preprocess, 19.4ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 21.9ms

Speed: 0.9ms preprocess, 21.9ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 23.1ms

Speed: 1.1ms preprocess, 23.1ms inference, 0.3ms postprocess per image at shape

(1, 3, 480, 640)

0: 448x640 2 license_plates, 22.2ms

Speed: 1.0ms preprocess, 22.2ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 20.1ms

Speed: 0.9ms preprocess, 20.1ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 18.9ms

Speed: 0.9ms preprocess, 18.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 19.7ms

Speed: 1.1ms preprocess, 19.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 416x640 1 license_plate, 19.3ms

Speed: 1.0ms preprocess, 19.3ms inference, 0.3ms postprocess per image at shape
(1, 3, 416, 640)

0: 480x640 1 license_plate, 19.5ms

Speed: 0.9ms preprocess, 19.5ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 20.1ms

Speed: 1.3ms preprocess, 20.1ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 19.0ms

Speed: 1.0ms preprocess, 19.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 416x640 4 license_plates, 20.8ms

Speed: 0.8ms preprocess, 20.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 416, 640)

0: 448x640 1 license_plate, 20.2ms

Speed: 0.9ms preprocess, 20.2ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 576x640 1 license_plate, 23.4ms

Speed: 1.2ms preprocess, 23.4ms inference, 0.3ms postprocess per image at shape
(1, 3, 576, 640)

0: 448x640 2 license_plates, 20.8ms

Speed: 1.0ms preprocess, 20.8ms inference, 0.3ms postprocess per image at shape

(1, 3, 448, 640)

0: 480x640 2 license_plates, 20.8ms

Speed: 1.0ms preprocess, 20.8ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 640x640 1 license_plate, 26.1ms

Speed: 1.2ms preprocess, 26.1ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 448x640 1 license_plate, 21.1ms

Speed: 0.9ms preprocess, 21.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 20.3ms

Speed: 0.9ms preprocess, 20.3ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 640x448 1 license_plate, 20.9ms

Speed: 1.0ms preprocess, 20.9ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 448)

0: 480x640 2 license_plates, 20.4ms

Speed: 1.0ms preprocess, 20.4ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 18.6ms

Speed: 1.0ms preprocess, 18.6ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 640x480 1 license_plate, 20.2ms

Speed: 1.1ms preprocess, 20.2ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 480x640 1 license_plate, 19.2ms

Speed: 1.0ms preprocess, 19.2ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 18.5ms

Speed: 1.0ms preprocess, 18.5ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 19.7ms

Speed: 1.0ms preprocess, 19.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 18.4ms

Speed: 1.1ms preprocess, 18.4ms inference, 0.3ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.7ms

Speed: 1.0ms preprocess, 18.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 21.3ms

Speed: 1.0ms preprocess, 21.3ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 20.5ms

Speed: 0.9ms preprocess, 20.5ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 19.3ms

Speed: 0.8ms preprocess, 19.3ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 20.7ms

Speed: 1.0ms preprocess, 20.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 20.0ms

Speed: 1.0ms preprocess, 20.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 19.3ms

Speed: 0.9ms preprocess, 19.3ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 19.7ms

Speed: 0.9ms preprocess, 19.7ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 20.4ms

Speed: 0.9ms preprocess, 20.4ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 23.0ms

Speed: 1.0ms preprocess, 23.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 3 license_plates, 20.9ms

Speed: 1.0ms preprocess, 20.9ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 21.8ms

Speed: 1.0ms preprocess, 21.8ms inference, 0.3ms postprocess per image at shape

(1, 3, 480, 640)

0: 640x640 1 license_plate, 25.3ms

Speed: 1.2ms preprocess, 25.3ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 480x640 2 license_plates, 19.0ms

Speed: 0.9ms preprocess, 19.0ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 2 license_plates, 21.0ms

Speed: 1.1ms preprocess, 21.0ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 20.3ms

Speed: 1.0ms preprocess, 20.3ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 2 license_plates, 21.6ms

Speed: 1.0ms preprocess, 21.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 20.2ms

Speed: 0.9ms preprocess, 20.2ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 18.7ms

Speed: 1.0ms preprocess, 18.7ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 640x640 1 license_plate, 29.9ms

Speed: 1.2ms preprocess, 29.9ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 640x640 1 license_plate, 25.0ms

Speed: 1.2ms preprocess, 25.0ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 480x640 1 license_plate, 19.4ms

Speed: 1.0ms preprocess, 19.4ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 512x640 1 license_plate, 20.0ms

Speed: 1.1ms preprocess, 20.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 512, 640)

0: 480x640 1 license_plate, 20.7ms

Speed: 1.0ms preprocess, 20.7ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 2 license_plates, 20.9ms

Speed: 0.9ms preprocess, 20.9ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 384x640 1 license_plate, 17.6ms

Speed: 0.8ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 480x640 1 license_plate, 18.8ms

Speed: 1.0ms preprocess, 18.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 19.6ms

Speed: 0.9ms preprocess, 19.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 18.0ms

Speed: 0.8ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 384x640 1 license_plate, 17.9ms

Speed: 0.7ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 384x640 1 license_plate, 16.1ms

Speed: 0.8ms preprocess, 16.1ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 480x640 2 license_plates, 18.8ms

Speed: 0.9ms preprocess, 18.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 19.7ms

Speed: 0.9ms preprocess, 19.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 18.5ms

Speed: 1.1ms preprocess, 18.5ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.4ms

Speed: 1.1ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 19.4ms

Speed: 0.9ms preprocess, 19.4ms inference, 0.3ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 19.8ms

Speed: 0.9ms preprocess, 19.8ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 640x480 1 license_plate, 21.6ms

Speed: 0.9ms preprocess, 21.6ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 448x640 (no detections), 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.1ms postprocess per image at shape (1, 3, 448, 640)

0: 640x640 1 license_plate, 24.9ms

Speed: 1.4ms preprocess, 24.9ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 480x640 1 license_plate, 18.5ms

Speed: 0.9ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 18.5ms

Speed: 0.9ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 (no detections), 18.5ms

Speed: 1.0ms preprocess, 18.5ms inference, 0.1ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 19.3ms

Speed: 1.0ms preprocess, 19.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 17.8ms

Speed: 1.0ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 640x640 1 license_plate, 23.5ms

Speed: 1.2ms preprocess, 23.5ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 480x640 1 license_plate, 18.6ms

Speed: 0.9ms preprocess, 18.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 19.9ms

Speed: 1.0ms preprocess, 19.9ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 19.5ms

Speed: 1.0ms preprocess, 19.5ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 19.4ms

Speed: 0.9ms preprocess, 19.4ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 1.0ms preprocess, 18.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 18.5ms

Speed: 1.0ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 16.2ms

Speed: 0.9ms preprocess, 16.2ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 448x640 1 license_plate, 19.0ms

Speed: 0.9ms preprocess, 19.0ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 3 license_plates, 16.4ms

Speed: 0.9ms preprocess, 16.4ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.5ms

Speed: 0.9ms preprocess, 17.5ms inference, 0.2ms postprocess per image at shape

(1, 3, 448, 640)

0: 448x640 1 license_plate, 17.4ms
Speed: 0.9ms preprocess, 17.4ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 19.0ms
Speed: 1.0ms preprocess, 19.0ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.5ms
Speed: 0.9ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 16.4ms
Speed: 0.9ms preprocess, 16.4ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 640x640 1 license_plate, 22.9ms
Speed: 1.2ms preprocess, 22.9ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 448x640 1 license_plate, 17.6ms
Speed: 1.0ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 16.5ms
Speed: 0.9ms preprocess, 16.5ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 18.3ms
Speed: 1.0ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 17.6ms
Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.6ms
Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 640x640 1 license_plate, 23.4ms
Speed: 1.2ms preprocess, 23.4ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x480 2 license_plates, 18.4ms
Speed: 1.0ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape

(1, 3, 640, 480)

0: 448x640 1 license_plate, 17.6ms
Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.9ms
Speed: 0.9ms preprocess, 17.9ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 17.8ms
Speed: 1.0ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 17.6ms
Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 20.1ms
Speed: 1.0ms preprocess, 20.1ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 544x640 2 license_plates, 20.3ms
Speed: 1.0ms preprocess, 20.3ms inference, 0.2ms postprocess per image at shape (1, 3, 544, 640)

0: 480x640 2 license_plates, 18.8ms
Speed: 0.9ms preprocess, 18.8ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 288x640 1 license_plate, 13.2ms
Speed: 0.7ms preprocess, 13.2ms inference, 0.2ms postprocess per image at shape (1, 3, 288, 640)

0: 448x640 1 license_plate, 18.3ms
Speed: 0.9ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 18.0ms
Speed: 1.0ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 19.9ms
Speed: 1.0ms preprocess, 19.9ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 17.4ms
Speed: 1.0ms preprocess, 17.4ms inference, 0.2ms postprocess per image at shape

(1, 3, 384, 640)

0: 448x640 1 license_plate, 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.3ms

Speed: 1.0ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 2 license_plates, 17.9ms

Speed: 1.0ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 3 license_plates, 17.3ms

Speed: 0.9ms preprocess, 17.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 17.9ms

Speed: 0.9ms preprocess, 17.9ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 3 license_plates, 17.8ms

Speed: 1.0ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 2 license_plates, 17.6ms

Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 23.3ms

Speed: 0.9ms preprocess, 23.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 512x640 1 license_plate, 20.0ms

Speed: 1.0ms preprocess, 20.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 512, 640)

0: 480x640 2 license_plates, 18.4ms

Speed: 0.9ms preprocess, 18.4ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.7ms

Speed: 1.0ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 640x480 2 license_plates, 18.5ms

Speed: 1.0ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape

(1, 3, 640, 480)

0: 448x640 2 license_plates, 18.6ms

Speed: 0.9ms preprocess, 18.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 8 license_plates, 17.1ms

Speed: 1.0ms preprocess, 17.1ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 2 license_plates, 17.5ms

Speed: 1.1ms preprocess, 17.5ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 640x640 1 license_plate, 22.5ms

Speed: 1.4ms preprocess, 22.5ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 512x640 1 license_plate, 20.1ms

Speed: 1.0ms preprocess, 20.1ms inference, 0.3ms postprocess per image at shape
(1, 3, 512, 640)

0: 608x640 1 license_plate, 21.7ms

Speed: 1.1ms preprocess, 21.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 608, 640)

0: 448x640 1 license_plate, 16.7ms

Speed: 1.1ms preprocess, 16.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 2 license_plates, 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 2 license_plates, 17.6ms

Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 640x480 2 license_plates, 18.2ms

Speed: 1.1ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape
(1, 3, 640, 480)

0: 480x640 1 license_plate, 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.5ms

Speed: 1.0ms preprocess, 17.5ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 5 license_plates, 18.3ms

Speed: 1.0ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 17.6ms

Speed: 1.0ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 16.6ms

Speed: 0.8ms preprocess, 16.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.1ms

Speed: 1.1ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 2 license_plates, 17.5ms

Speed: 0.9ms preprocess, 17.5ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 2 license_plates, 21.3ms

Speed: 0.9ms preprocess, 21.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 512x640 1 license_plate, 18.9ms

Speed: 0.9ms preprocess, 18.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 512, 640)

0: 384x640 3 license_plates, 15.8ms

Speed: 0.7ms preprocess, 15.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 480x640 2 license_plates, 18.2ms

Speed: 1.0ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 352x640 2 license_plates, 14.9ms

Speed: 0.7ms preprocess, 14.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 352, 640)

0: 384x640 2 license_plates, 15.6ms

Speed: 0.8ms preprocess, 15.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 480x640 2 license_plates, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.4ms

Speed: 1.0ms preprocess, 18.4ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 (no detections), 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.1ms postprocess per image at shape (1, 3, 480, 640)

0: 640x640 1 license_plate, 22.7ms

Speed: 1.2ms preprocess, 22.7ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 480x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 640x640 1 license_plate, 22.4ms

Speed: 1.2ms preprocess, 22.4ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 448x640 2 license_plates, 17.2ms

Speed: 0.9ms preprocess, 17.2ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 640x480 2 license_plates, 17.7ms

Speed: 1.0ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 480x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 3 license_plates, 18.0ms

Speed: 0.9ms preprocess, 18.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 416x640 1 license_plate, 16.6ms

Speed: 0.8ms preprocess, 16.6ms inference, 0.2ms postprocess per image at shape (1, 3, 416, 640)

0: 480x640 2 license_plates, 18.0ms

Speed: 1.0ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 18.5ms

Speed: 0.9ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape

(1, 3, 448, 640)

0: 640x640 1 license_plate, 23.6ms

Speed: 1.2ms preprocess, 23.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 320x640 1 license_plate, 16.3ms

Speed: 0.7ms preprocess, 16.3ms inference, 0.3ms postprocess per image at shape
(1, 3, 320, 640)

0: 576x640 1 license_plate, 21.3ms

Speed: 1.1ms preprocess, 21.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 576, 640)

0: 640x640 1 license_plate, 32.6ms

Speed: 1.2ms preprocess, 32.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 512x640 2 license_plates, 18.6ms

Speed: 0.9ms preprocess, 18.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 512, 640)

0: 480x640 2 license_plates, 17.9ms

Speed: 1.0ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 17.6ms

Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 2 license_plates, 18.5ms

Speed: 1.0ms preprocess, 18.5ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 1.0ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 0.9ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 416x640 2 license_plates, 16.0ms

Speed: 0.9ms preprocess, 16.0ms inference, 0.2ms postprocess per image at shape

(1, 3, 416, 640)

0: 480x640 1 license_plate, 18.6ms

Speed: 1.0ms preprocess, 18.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 16.6ms

Speed: 0.9ms preprocess, 16.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 16.6ms

Speed: 0.8ms preprocess, 16.6ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 544x640 3 license_plates, 18.8ms

Speed: 1.1ms preprocess, 18.8ms inference, 0.2ms postprocess per image at shape (1, 3, 544, 640)

0: 480x640 2 license_plates, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 18.1ms

Speed: 0.8ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 384x640 3 license_plates, 16.6ms

Speed: 0.8ms preprocess, 16.6ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 448x640 1 license_plate, 17.2ms

Speed: 0.9ms preprocess, 17.2ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 512x640 2 license_plates, 19.7ms

Speed: 1.0ms preprocess, 19.7ms inference, 0.2ms postprocess per image at shape

(1, 3, 512, 640)

0: 480x640 3 license_plates, 18.4ms

Speed: 0.9ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.9ms

Speed: 1.0ms preprocess, 18.9ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 16.9ms

Speed: 0.9ms preprocess, 16.9ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 1.0ms preprocess, 18.0ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 384x640 1 license_plate, 15.7ms

Speed: 0.9ms preprocess, 15.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 448x640 1 license_plate, 18.5ms

Speed: 0.9ms preprocess, 18.5ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 2 license_plates, 17.5ms

Speed: 0.9ms preprocess, 17.5ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.8ms

Speed: 1.0ms preprocess, 17.8ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.6ms

Speed: 0.9ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 17.2ms

Speed: 1.0ms preprocess, 17.2ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 640x640 1 license_plate, 24.7ms

Speed: 1.2ms preprocess, 24.7ms inference, 0.3ms postprocess per image at shape
(1, 3, 640, 640)

0: 448x640 1 license_plate, 16.8ms

Speed: 0.8ms preprocess, 16.8ms inference, 0.2ms postprocess per image at shape

(1, 3, 448, 640)

0: 480x640 2 license_plates, 17.6ms

Speed: 1.0ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 15.7ms

Speed: 0.8ms preprocess, 15.7ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 480x640 1 license_plate, 17.7ms

Speed: 1.0ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 640x640 1 license_plate, 25.9ms

Speed: 1.2ms preprocess, 25.9ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 0.9ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 608x640 2 license_plates, 21.7ms

Speed: 1.2ms preprocess, 21.7ms inference, 0.2ms postprocess per image at shape (1, 3, 608, 640)

0: 480x640 1 license_plate, 17.6ms

Speed: 1.0ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 576x640 1 license_plate, 20.4ms

Speed: 1.0ms preprocess, 20.4ms inference, 0.2ms postprocess per image at shape (1, 3, 576, 640)

0: 448x640 1 license_plate, 17.7ms

Speed: 1.0ms preprocess, 17.7ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 16.8ms

Speed: 0.9ms preprocess, 16.8ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.8ms

Speed: 1.0ms preprocess, 17.8ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 17.6ms

Speed: 0.9ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.0ms

Speed: 0.9ms preprocess, 17.0ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 640x640 3 license_plates, 22.6ms

Speed: 1.2ms preprocess, 22.6ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 448x640 1 license_plate, 17.0ms

Speed: 0.9ms preprocess, 17.0ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 18.3ms

Speed: 0.9ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.2ms

Speed: 1.0ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 640x480 2 license_plates, 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 480x640 (no detections), 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.1ms postprocess per image at shape (1, 3, 480, 640)

0: 640x416 (no detections), 17.2ms

Speed: 0.8ms preprocess, 17.2ms inference, 0.1ms postprocess per image at shape (1, 3, 640, 416)

0: 448x640 1 license_plate, 17.0ms

Speed: 0.9ms preprocess, 17.0ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 512x640 1 license_plate, 21.3ms

Speed: 0.9ms preprocess, 21.3ms inference, 0.2ms postprocess per image at shape

(1, 3, 512, 640)

0: 480x640 1 license_plate, 17.7ms

Speed: 1.0ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 2 license_plates, 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 17.3ms

Speed: 0.9ms preprocess, 17.3ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 17.6ms

Speed: 0.9ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 17.8ms

Speed: 0.9ms preprocess, 17.8ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 480x640 1 license_plate, 19.1ms

Speed: 1.0ms preprocess, 19.1ms inference, 0.3ms postprocess per image at shape
(1, 3, 480, 640)

0: 448x640 1 license_plate, 19.0ms

Speed: 0.9ms preprocess, 19.0ms inference, 0.3ms postprocess per image at shape
(1, 3, 448, 640)

0: 384x640 2 license_plates, 16.0ms

Speed: 0.8ms preprocess, 16.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 384, 640)

0: 448x640 1 license_plate, 16.6ms

Speed: 0.9ms preprocess, 16.6ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 448x640 1 license_plate, 17.0ms

Speed: 0.9ms preprocess, 17.0ms inference, 0.2ms postprocess per image at shape
(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.3ms

Speed: 1.0ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 640x640 1 license_plate, 23.6ms
Speed: 1.4ms preprocess, 23.6ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 384x640 4 license_plates, 15.8ms
Speed: 0.7ms preprocess, 15.8ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 448x640 1 license_plate, 17.7ms
Speed: 0.9ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 640x480 2 license_plates, 18.0ms
Speed: 0.9ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 448x640 1 license_plate, 17.3ms
Speed: 0.9ms preprocess, 17.3ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.1ms
Speed: 0.9ms preprocess, 17.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 19.3ms
Speed: 0.9ms preprocess, 19.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 16.5ms
Speed: 1.0ms preprocess, 16.5ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 18.9ms
Speed: 1.0ms preprocess, 18.9ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 17.8ms
Speed: 0.9ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.5ms
Speed: 1.0ms preprocess, 17.5ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 22.4ms
Speed: 0.9ms preprocess, 22.4ms inference, 0.2ms postprocess per image at shape

(1, 3, 448, 640)

0: 448x640 2 license_plates, 22.0ms

Speed: 1.1ms preprocess, 22.0ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 23.1ms

Speed: 0.9ms preprocess, 23.1ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 24.4ms

Speed: 0.9ms preprocess, 24.4ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 320x640 1 license_plate, 21.7ms

Speed: 0.9ms preprocess, 21.7ms inference, 0.2ms postprocess per image at shape (1, 3, 320, 640)

0: 384x640 1 license_plate, 19.5ms

Speed: 0.8ms preprocess, 19.5ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 448x640 2 license_plates, 17.4ms

Speed: 0.9ms preprocess, 17.4ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 16.9ms

Speed: 0.9ms preprocess, 16.9ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 384x640 2 license_plates, 16.0ms

Speed: 0.8ms preprocess, 16.0ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 448x640 1 license_plate, 17.4ms

Speed: 0.9ms preprocess, 17.4ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 17.9ms

Speed: 1.0ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.1ms

Speed: 0.9ms preprocess, 17.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.4ms

Speed: 0.9ms preprocess, 17.4ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 544x640 1 license_plate, 20.4ms

Speed: 1.0ms preprocess, 20.4ms inference, 0.2ms postprocess per image at shape (1, 3, 544, 640)

0: 448x640 1 license_plate, 19.2ms

Speed: 0.9ms preprocess, 19.2ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 384x640 1 license_plate, 15.4ms

Speed: 1.0ms preprocess, 15.4ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 512x640 1 license_plate, 19.0ms

Speed: 1.0ms preprocess, 19.0ms inference, 0.3ms postprocess per image at shape (1, 3, 512, 640)

0: 480x640 1 license_plate, 18.7ms

Speed: 1.0ms preprocess, 18.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 576x640 1 license_plate, 21.0ms

Speed: 1.1ms preprocess, 21.0ms inference, 0.2ms postprocess per image at shape (1, 3, 576, 640)

0: 480x640 1 license_plate, 18.8ms

Speed: 0.9ms preprocess, 18.8ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 352x640 1 license_plate, 14.5ms

Speed: 0.7ms preprocess, 14.5ms inference, 0.2ms postprocess per image at shape (1, 3, 352, 640)

0: 448x640 (no detections), 17.1ms

Speed: 0.8ms preprocess, 17.1ms inference, 0.1ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 17.6ms

Speed: 1.0ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 15.9ms

Speed: 0.7ms preprocess, 15.9ms inference, 0.3ms postprocess per image at shape (1, 3, 384, 640)

0: 352x640 1 license_plate, 14.5ms

Speed: 0.7ms preprocess, 14.5ms inference, 0.2ms postprocess per image at shape

(1, 3, 352, 640)

0: 384x640 2 license_plates, 16.0ms

Speed: 0.8ms preprocess, 16.0ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 480x640 1 license_plate, 19.2ms

Speed: 1.0ms preprocess, 19.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.3ms

Speed: 1.0ms preprocess, 18.3ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.1ms

Speed: 0.8ms preprocess, 17.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.8ms

Speed: 1.0ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.1ms

Speed: 0.9ms preprocess, 17.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 16.9ms

Speed: 0.9ms preprocess, 16.9ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.9ms

Speed: 1.0ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 19.0ms

Speed: 0.9ms preprocess, 19.0ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 2 license_plates, 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 512x640 2 license_plates, 20.2ms

Speed: 1.1ms preprocess, 20.2ms inference, 0.2ms postprocess per image at shape (1, 3, 512, 640)

0: 480x640 1 license_plate, 17.9ms

Speed: 1.0ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 448x640 1 license_plate, 17.0ms
Speed: 0.9ms preprocess, 17.0ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 384x640 1 license_plate, 16.1ms
Speed: 0.8ms preprocess, 16.1ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 480x640 1 license_plate, 17.6ms
Speed: 0.9ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 17.5ms
Speed: 0.9ms preprocess, 17.5ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.2ms
Speed: 0.8ms preprocess, 17.2ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.6ms
Speed: 1.0ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 18.4ms
Speed: 0.9ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 19.8ms
Speed: 1.0ms preprocess, 19.8ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 640x480 1 license_plate, 19.4ms
Speed: 1.0ms preprocess, 19.4ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 480)

0: 448x640 1 license_plate, 18.1ms
Speed: 0.9ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 640x640 1 license_plate, 24.9ms
Speed: 1.2ms preprocess, 24.9ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 544x640 2 license_plates, 20.6ms
Speed: 1.0ms preprocess, 20.6ms inference, 0.2ms postprocess per image at shape

(1, 3, 544, 640)

0: 480x640 1 license_plate, 19.5ms

Speed: 1.0ms preprocess, 19.5ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.3ms

Speed: 0.9ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.1ms

Speed: 0.8ms preprocess, 17.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.6ms

Speed: 1.1ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 640x512 3 license_plates, 21.0ms

Speed: 1.0ms preprocess, 21.0ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 512)

0: 640x416 2 license_plates, 16.3ms

Speed: 0.9ms preprocess, 16.3ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 416)

0: 480x640 1 license_plate, 18.3ms

Speed: 0.9ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 15.6ms

Speed: 0.8ms preprocess, 15.6ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 384x640 1 license_plate, 15.1ms

Speed: 0.8ms preprocess, 15.1ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 0.9ms preprocess, 18.0ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 512x640 1 license_plate, 19.1ms

Speed: 1.0ms preprocess, 19.1ms inference, 0.2ms postprocess per image at shape

(1, 3, 512, 640)

0: 448x640 1 license_plate, 18.1ms

Speed: 1.0ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 (no detections), 17.7ms

Speed: 1.0ms preprocess, 17.7ms inference, 0.1ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.2ms

Speed: 0.9ms preprocess, 17.2ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 17.7ms

Speed: 0.9ms preprocess, 17.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 576x640 2 license_plates, 20.3ms

Speed: 1.1ms preprocess, 20.3ms inference, 0.2ms postprocess per image at shape (1, 3, 576, 640)

0: 480x640 1 license_plate, 18.1ms

Speed: 0.9ms preprocess, 18.1ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.9ms

Speed: 0.9ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.1ms

Speed: 1.0ms preprocess, 17.1ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 352x640 2 license_plates, 15.3ms

Speed: 0.7ms preprocess, 15.3ms inference, 0.2ms postprocess per image at shape (1, 3, 352, 640)

0: 640x544 1 license_plate, 19.7ms

Speed: 1.0ms preprocess, 19.7ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 544)

0: 480x640 1 license_plate, 18.3ms

Speed: 1.0ms preprocess, 18.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 17.5ms

Speed: 1.0ms preprocess, 17.5ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 448x640 1 license_plate, 18.9ms
Speed: 0.9ms preprocess, 18.9ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 18.7ms
Speed: 0.9ms preprocess, 18.7ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 640x640 1 license_plate, 23.2ms
Speed: 1.3ms preprocess, 23.2ms inference, 0.2ms postprocess per image at shape (1, 3, 640, 640)

0: 480x640 1 license_plate, 17.6ms
Speed: 1.1ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 15.8ms
Speed: 0.7ms preprocess, 15.8ms inference, 0.2ms postprocess per image at shape (1, 3, 384, 640)

0: 480x640 1 license_plate, 18.4ms
Speed: 0.9ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.6ms
Speed: 0.9ms preprocess, 18.6ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.6ms
Speed: 0.8ms preprocess, 17.6ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 1 license_plate, 17.6ms
Speed: 1.0ms preprocess, 17.6ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 512x640 2 license_plates, 19.4ms
Speed: 1.0ms preprocess, 19.4ms inference, 0.3ms postprocess per image at shape (1, 3, 512, 640)

0: 640x640 1 license_plate, 22.7ms
Speed: 1.3ms preprocess, 22.7ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 480x640 1 license_plate, 17.3ms
Speed: 1.0ms preprocess, 17.3ms inference, 0.2ms postprocess per image at shape

(1, 3, 480, 640)

0: 480x640 1 license_plate, 18.0ms

Speed: 1.0ms preprocess, 18.0ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 544x640 1 license_plate, 19.7ms

Speed: 1.1ms preprocess, 19.7ms inference, 0.3ms postprocess per image at shape (1, 3, 544, 640)

0: 480x640 2 license_plates, 17.8ms

Speed: 1.0ms preprocess, 17.8ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 352x640 1 license_plate, 14.5ms

Speed: 0.7ms preprocess, 14.5ms inference, 0.2ms postprocess per image at shape (1, 3, 352, 640)

0: 480x640 2 license_plates, 18.4ms

Speed: 0.9ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 2 license_plates, 17.5ms

Speed: 0.9ms preprocess, 17.5ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 640x640 1 license_plate, 25.7ms

Speed: 1.2ms preprocess, 25.7ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 2 license_plates, 23.6ms

Speed: 1.3ms preprocess, 23.6ms inference, 0.3ms postprocess per image at shape (1, 3, 640, 640)

0: 448x640 2 license_plates, 16.8ms

Speed: 0.8ms preprocess, 16.8ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 16.8ms

Speed: 0.9ms preprocess, 16.8ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 17.3ms

Speed: 0.8ms preprocess, 17.3ms inference, 0.2ms postprocess per image at shape

(1, 3, 448, 640)

0: 480x640 1 license_plate, 17.4ms

Speed: 0.9ms preprocess, 17.4ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 352x640 1 license_plate, 14.9ms

Speed: 0.9ms preprocess, 14.9ms inference, 0.2ms postprocess per image at shape (1, 3, 352, 640)

0: 480x640 (no detections), 18.5ms

Speed: 1.0ms preprocess, 18.5ms inference, 0.1ms postprocess per image at shape (1, 3, 480, 640)

0: 384x640 1 license_plate, 15.3ms

Speed: 0.8ms preprocess, 15.3ms inference, 0.3ms postprocess per image at shape (1, 3, 384, 640)

0: 480x640 3 license_plates, 24.1ms

Speed: 0.9ms preprocess, 24.1ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 18.4ms

Speed: 0.9ms preprocess, 18.4ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 480x640 2 license_plates, 19.3ms

Speed: 1.0ms preprocess, 19.3ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.8ms

Speed: 0.8ms preprocess, 17.8ms inference, 0.2ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 512x640 1 license_plate, 20.0ms

Speed: 0.9ms preprocess, 20.0ms inference, 0.2ms postprocess per image at shape (1, 3, 512, 640)

0: 480x640 1 license_plate, 19.1ms

Speed: 0.9ms preprocess, 19.1ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.9ms

Speed: 0.9ms preprocess, 17.9ms inference, 0.2ms postprocess per image at shape

(1, 3, 448, 640)

0: 480x640 1 license_plate, 18.2ms

Speed: 0.9ms preprocess, 18.2ms inference, 0.2ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 1 license_plate, 18.7ms

Speed: 1.0ms preprocess, 18.7ms inference, 0.3ms postprocess per image at shape (1, 3, 480, 640)

0: 448x640 1 license_plate, 17.1ms

Speed: 0.9ms preprocess, 17.1ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

0: 448x640 1 license_plate, 19.2ms

Speed: 0.9ms preprocess, 19.2ms inference, 0.3ms postprocess per image at shape (1, 3, 448, 640)

3 Measuring Evaluation Metrics

```
[4]: import time
import numpy as np

def evaluate_performance(model, test_images_path):
    latencies = []

    # Measure Inference Latency
    for img_path in test_images_path[:50]: # Sample 50 images
        start_time = time.time()
        model.predict(img_path, verbose=False)
        end_time = time.time()
        latencies.append((end_time - start_time) * 1000) # Convert to ms

    avg_latency = np.mean(latencies)
    print(f"--- Performance Report ---")
    print(f"Average Inference Latency: {avg_latency:.2f} ms per frame")
    print(f"Throughput: {1000/avg_latency:.2f} FPS")

# Run the evaluation
evaluate_performance(model, test_images)
```

--- Performance Report ---

Average Inference Latency: 22.16 ms per frame

Throughput: 45.12 FPS

4 Edge Deployment (Exporting)

```
[6]: import sys
      !{sys.executable} -m pip install onnx onnxslim
```

```
Collecting onnx
  Using cached onnx-1.19.1-cp39-cp39-macosx_12_0_universal2.whl.metadata (7.0
kB)
Collecting onnxslim
  Using cached onnxslim-0.1.87-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: numpy>=1.22 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
onnx) (2.0.2)
Requirement already satisfied: protobuf>=4.25.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
onnx) (6.33.5)
Requirement already satisfied: typing_extensions>=4.7.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
onnx) (4.15.0)
Collecting ml_dtypes>=0.5.0 (from onnx)
  Using cached ml_dtypes-0.5.4-cp39-cp39-macosx_10_9_universal2.whl.metadata
(8.9 kB)
Collecting colorama (from onnxslim)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: packaging in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
onnxslim) (26.0)
Requirement already satisfied: sympy>=1.13.1 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
onnxslim) (1.14.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/Users/kishankunal/workspace/AppliedAI/.venv/lib/python3.9/site-packages (from
sympy>=1.13.1->onnxslim) (1.3.0)
Using cached onnx-1.19.1-cp39-cp39-macosx_12_0_universal2.whl (18.3 MB)
Using cached onnxslim-0.1.87-py3-none-any.whl (235 kB)
Using cached ml_dtypes-0.5.4-cp39-cp39-macosx_10_9_universal2.whl (676 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Installing collected packages: ml_dtypes, colorama, onnx, onnxslim
      4/4

[onnxslim]2/4 [onnx]
Successfully installed colorama-0.4.6 ml_dtypes-0.5.4 onnx-1.19.1
onnxslim-0.1.87
```

```
[8]: path = model.export(format='onnx')
      print(f" Success! Your production-ready model is at: {path}")
```

Ultralytics 8.4.11 Python-3.9.6 torch-2.8.0 CPU (Apple M5 Pro)

```
PyTorch: starting from '/Users/kishankunal/workspace/AppliedAI/runs
/detect/plate_anonymizer_v13/weights/best.pt' with input shape (1, 3, 640, 640)
BCHW and output shape(s) (1, 5, 8400) (5.9 MB)
```

```
ONNX: starting export with onnx 1.19.1 opset 22...
```

```
ONNX: slimming with onnxslim 0.1.87...
```

```
ONNX: export success 0.4s, saved as '/Users/kishankunal/workspace
/AppliedAI/runs/detect/plate_anonymizer_v13/weights/best.onnx' (11.7 MB)
```

```
Export complete (0.4s)
```

```
Results saved to /Users/kishankunal/workspace/AppliedAI/runs/detect/plate_an
onymizer_v13/weights
```

```
Predict: yolo predict task=detect model=/Users/kishankunal/workspace/App
liedAI/runs/detect/plate_anonymizer_v13/weights/best.onnx imgsz=640
```

```
Validate: yolo val task=detect model=/Users/kishankunal/workspace/Applied
AI/runs/detect/plate_anonymizer_v13/weights/best.onnx imgsz=640
```

```
data=license_plate.yaml
```

```
Visualize: https://netron.app
```

```
Success! Your production-ready model is at: /Users/kishankunal/workspace/Appli
edAI/runs/detect/plate_anonymizer_v13/weights/best.onnx
```

5 Continuous Learning (Feedback Loop)

This script implements your recommendation for a Feedback Loop. It identifies “Low-Confidence” detections (e.g., between 25% and 50% confidence) and saves them to a specific folder for human review.

```
[9]: def feedback_loop_filter(img_path, model, review_folder="needs_review"):
    os.makedirs(review_folder, exist_ok=True)
    results = model(img_path)[0]

    for box in results.bboxes:
        conf = box.conf.item()
        # STRATEGIC LOGIC: Flag low-confidence detections for human review
        if 0.25 < conf < 0.50:
            print(f" Low confidence ({conf:.2f}) detected for {img_path}.␣
↪Flagging for review.")
            filename = os.path.basename(img_path)
            # Copy or save image to review folder
            cv2.imwrite(f"{review_folder}/review_{filename}", results.orig_img)
```

```
[10]: import os
import cv2
from pathlib import Path

# 1. Define Confidence Zones
```

```

CONFIDENCE_THRESHOLD = 0.50 # Anything above this is blurred automatically
REVIEW_ZONE_START = 0.25 # Anything between 0.25 and 0.50 goes to Review

# 2. Setup Folders
os.makedirs("output_redacted", exist_ok=True)
os.makedirs("needs_human_review", exist_ok=True)

def test_feedback_loop(image_list, model):
    for img_path in image_list:
        img = cv2.imread(str(img_path))
        results = model(img, verbose=False)[0]

        needs_review = False
        detections_found = False

        for box in results.bboxes:
            conf = box.conf.item()
            detections_found = True

            # CASE A: High Confidence -> Redact
            if conf >= CONFIDENCE_THRESHOLD:
                x1, y1, x2, y2 = map(int, box.xyxy[0])
                roi = img[y1:y2, x1:x2]
                img[y1:y2, x1:x2] = cv2.GaussianBlur(roi, (35, 35), 0)

            # CASE B: Low Confidence -> Flag for Feedback Loop
            elif REVIEW_ZONE_START <= conf < CONFIDENCE_THRESHOLD:
                needs_review = True

        # Save to appropriate folder
        filename = img_path.name
        if needs_review:
            cv2.imwrite(f"needs_human_review/REVIEW_{filename}", img)
            print(f" {filename}: Flagged for Human Review (Conf: {conf:.2f})")
        elif detections_found:
            cv2.imwrite(f"output_redacted/REDACTED_{filename}", img)
            print(f" {filename}: Successfully Redacted.")
        else:
            print(f" {filename}: No plates detected.")

# Run the test
test_images = list(Path("./data/images/test").glob("*.jpg"))
test_feedback_loop(test_images, model)

```

```

0f0596b1c511e071.jpg: Successfully Redacted.
791d63b1c3499e4f.jpg: Flagged for Human Review (Conf: 0.29)
5daafbcf76fa6602.jpg: Successfully Redacted.
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